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An Introduction To Auto-Callables

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Auto-callables have been actively traded in Europe for several years and have now become an increasingly influential force in flow derivatives. Initially, they were issued on baskets of stocks, then on indices and now increasingly on single stocks.

Recent activity in Asia, for example in Korea in January last year, suggests structured products play a pivotal role in volatility markets, although the theoretical effect in Europe and the U.S. is diluted because of greater activity from hedge funds, corporates, institutional investors and overwriting funds.

An auto-callable product is a structured product that has an automatic call feature on pre-prescribed dates known as the auto-call dates. These are usually from monthly to bi-annual intervals.

While few auto-callables are exact

1. A large coupon is payable if an a the strike on an auto-call date. If no for the next auto-call date; and

2. A sold knock-in put is often emb protection because the product is

For illustrative purposes in this Lea

- Three-year maturity with annual

- 10% coupon escalating to 20% and 30% in years two and three if not auto-called; and

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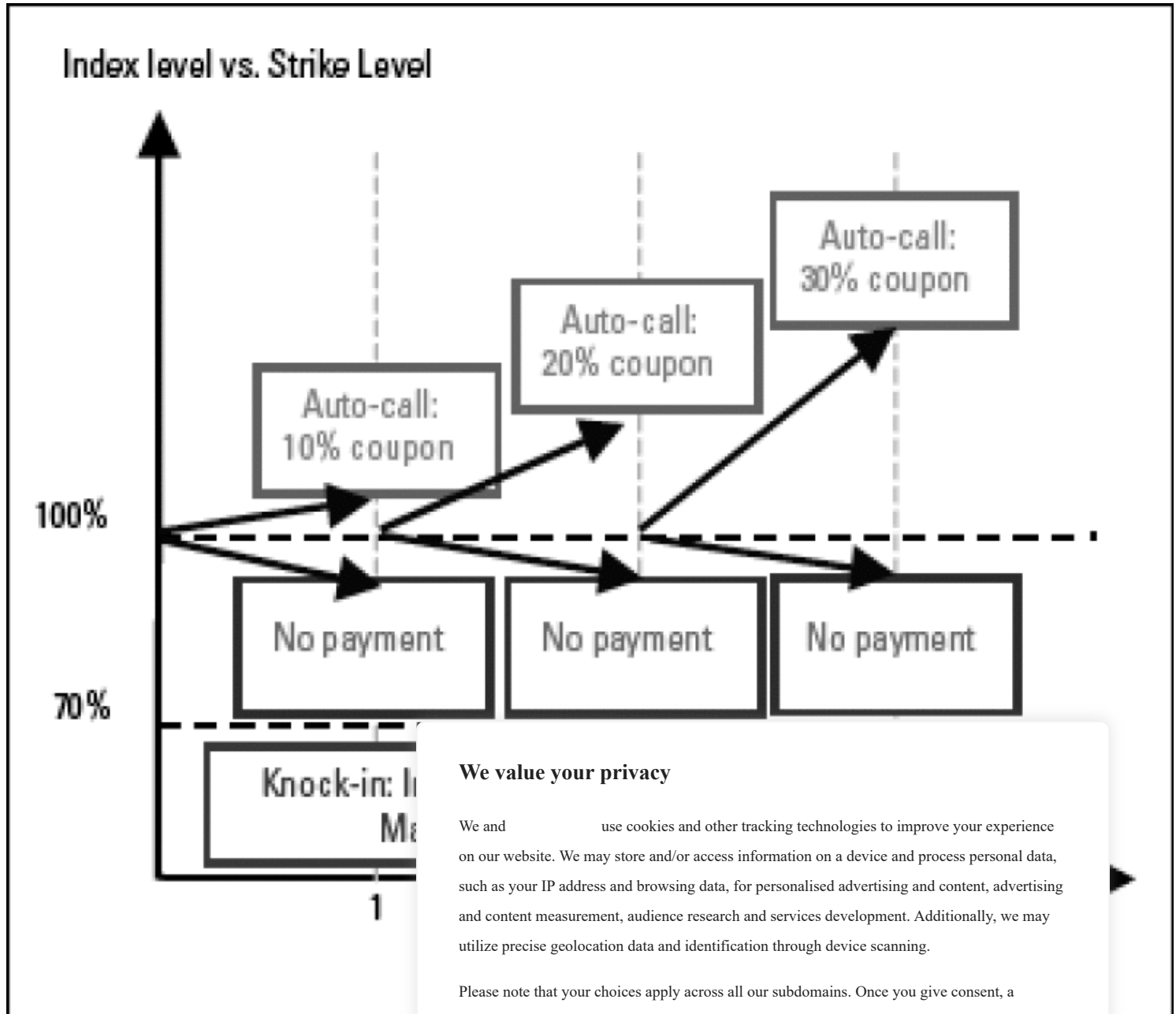
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- 70% downside knock-in put with continuous barrier which can knock-in any time.

At each auto-call date there are two possibilities that determine if the product will be auto-called or will not pay a coupon but continue to the next period:

- If the underlying is above the strike (100%), accumulated coupons are paid at 10% per year of life and the whole structure is terminated; and



- If the underlying is below the strike level, the structure escalates by a further 10% for the next period.

If at any time the underlying is below the strike level, the structure becomes short the put and this structure is terminated.

Technically speaking, in an auto-callable product the client is short an out-of-the-money knock-in put and long a strip of contingent at-the-money digital options, which pay coupons and knock out the whole product if it is in the money at the auto-call dates.

Product Popularity

The most common explanation for auto-callable popularity is the potential for a high coupon, which is attractive in the current low yield environment for bonds, property and equities.

Investors who buy auto-callable products are looking for a long-term equity-linked investment providing a conditional 100% principal protection at maturity and a potential for a large return on investment from coupons. The investor has a view the underlying will have a moderate increase over the life of the product and will not trade below the knock-in level.

Another source of popularity arises from the fact auto-callable products may have a shorter lifespan than other structured products. It is hoped in these situations, investors flush with new cash from the coupon paid, might invest in further auto-call products. Many other structured products lock in an investor's money for a long duration--five to 10 years is common--so for issuers, the prospect of a seemingly long-term product expiring early may provide the chance to issue another product to the same investor.

Auto-Call Risks

From an investor's perspective the risks are:

- * Early maturity;
- * Credit;
- * Tax;
- * Performance of the underlying--sold knock-in strike;
- * A rise in interest rates during the life of the product;
- * Liquidity risk and spread costs if the product is not actively traded;
- * Event risk due to some adjustments to the terms of the contract such as mergers and trading suspension.

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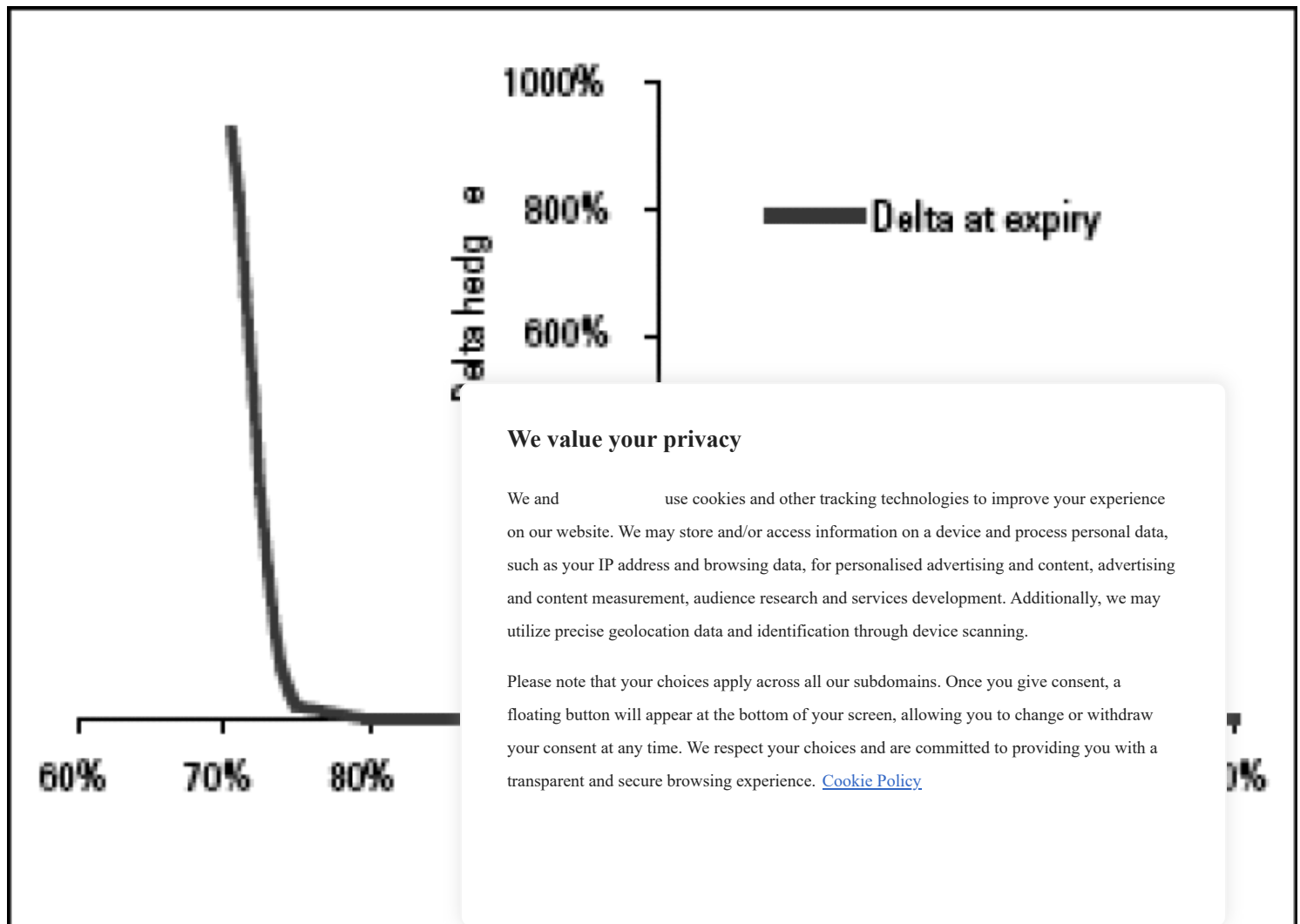
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From an issuer's perspective the risks are:

- * A discontinuous valuation pattern near auto-call dates when delta hedging may require a theoretically infinite multiple of the notional to be traded rapidly;
- * An indeterminate time to maturity. Vega and gamma exposure is highly dynamic. As the underlying rallies, vega exposure gravitates toward the next auto-call date; and
- * Correlation exposure between bonds and the underlying. The sensitivity of the fair value of an auto-call product to this hybrid correlation is negative. This stems from the risk to the issuer that as the underlying rallies, the value of the auto-callable increases but if bonds are also rallying, the discount rate falls and will also serve to increase the product's fair value. This is because the present value of the coupon payment will be higher. In this situation, high correlation has a double negative effect for the issuer.

Auto-Call Effect On Flow Derivative Markets



Delta

The delta changes can be significant and irregular for auto-call products particularly when the underlying is trading near a knock-in or knock-out level close to expiry.

The auto-callable product experiences a fair value jump when a knock-in or knock-out occurs. In our example, at the 70% level, the issuer requires a large hedge position so that profit/loss on minor underlying movements are sufficient to offset large fair value changes in the auto-callable.

The chart shows the massive change in delta position required when the underlying of an auto-callable trades close to a knock-in or knock-out level near an auto-call date or expiry.

Vega

The auto-callable issuer is buying volatility. When the underlying is near a level at which a coupon would be payable, the issuer would prefer the stock to be highly volatile to provide a greater probable distance from the barrier.

If the underlying declines, the issuer still prefers high volatility to provide a greater probability of hitting the knock-in level when the issuer becomes long a put option.

This long volatility exposure is hedged by issuers selling volatility. The indeterminate maturity, however, makes this hedge highly dynamic and pressures the term structure and skew when flows are significant.

Term Structure

* As the underlying rallies, an auto-call product shortens. Longer dated ve hedges. The opposite applies for a

Skews

* Similar to the effect on term struc from issuers increases for upside o the knock-in put being triggered in advance. Both of these effects ser

Gamma

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Gamma wells are an interesting effect experienced by issuers of auto-callables. These are created when one or more institutions hold large derivative positions and the required hedging activity on the underlying asset begin to influence its price movements.

Long gamma means delta hedging pressure tends to dampen volatility and sometimes supports the share price near or above the strike price near maturity.

At the knock-in level and especially close to an auto-call date or expiry, the gamma becomes large, which means dealers buy more and more of the underlying on the way down, providing support for the market.

The caveat is that at some point, the issuer will consider the release of their delta position to be more important than delta hedging profits and gamma becomes irrelevant because the issuing bank sells its hedge. In practice, few dealers will fully hedge themselves when the market approaches the barrier and will often sell off the delta before the barrier is breached in an attempt to achieve the anticipated correct hedge ratio.

Auto-callables are significant, but vary widely in maturity and strikes making gamma wells at any particular underlying level or on any particular date relatively small compared to other market forces at this stage, in our view. At the current low implied volatility levels, even auto-callables are hard to price with significant yield. Demand may subside if volatility remains low and bond yields rise further.

*This week's Learning Curve was written by **Gerry Fowler CFA** and **Lamia Outgenza**, equity derivative strategists at **Citigroup** in London.*

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